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Designed for discourse: how reactions and replies create trade-offs in political participation

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Author note: The data for this project were collected when Harshit Aneja and Yiting Chen were affiliated with the National University of Singapore.

Abstract

While social media platforms aim to foster quality discussions through interactive features, existing theories often overlook how these features may produce divergent outcomes. We build on the Theory of Interactive Media Effects model to study how interactive features affect emergent user behavior and interaction patterns. Through an experiment ($N = 1{,}370$) on a custom-built platform, we tested how two common features—reactions and replies—influence participation, engagement, and diversity in political discussions. Our findings reveal a fundamental trade-off between these features. Reaction features, which are inherently low-effort, enabled quick responses but reduced overall message volume, while simultaneously increasing the elaboration of individual posts and exposure to diverse viewpoints. In contrast, reply features, which are inherently high-effort, supported targeted exchanges but deepened interactions within narrow threads, fragmenting conversation and reducing group-level diversity. Our findings demonstrate that while interactive features may facilitate interaction, they structure trade-offs between different forms of inclusiveness, thereby offering new insights for theorizing and designing more equitable platform mechanics.

Lay Summary

In this study, we tested how two common social media features—reaction buttons (such as upvotes and downvotes) and reply tools—shape political discussions online. We built a custom chat platform and recruited over 1,300 participants to discuss a political topic in small groups, randomly assigning each group to use different combinations of features. We found that when reaction buttons were available, people posted fewer messages overall, but the messages they wrote were longer. Reactions also exposed users to a broader range of viewpoints. However, people tended to cluster around popular topics, likely in an effort to earn more upvotes. When reply features were available, conversations ran deeper, with longer back-and-forth threads. However, replies narrowed the range of opinions expressed, as discussions became anchored around fewer viewpoints. Together, these findings suggest that no single feature improves online political discussions across the board. Instead, reactions and replies create real trade-offs between participation, engagement, and the diversity of perspectives. We suggest that platform designers must carefully consider which goals matter most, as choices that support one outcome may come at the expense of another.

Inclusiveness is a core attribute of high-quality political discussion, reflecting not only who participates but also how they engage and whether diverse perspectives are sustained. On social media, inclusiveness is shaped by design features that govern interaction. Seemingly minor design choices encourage different forms of social behavior; for instance, platforms such as Twitter/X and Reddit emphasize amplification, while Discord emphasizes conversational exchange (Esau et al., 2017; Neubaum & Weeks, 2023). Recent work demonstrates how such features shape rhetorical strategies in identity-centric debates (Pyle et al., 2025), resulting in distinct participation ecosystems. Yet research has only begun to probe how design features enable or hinder inclusiveness.

A growing shift in platform design makes this feature-level perspective increasingly consequential. While public broadcast feeds remain central, contemporary social media platforms increasingly

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structure participation and engagement around conversational and semi-private interaction spaces in which communication unfolds through replies, reactions, and small-group exchanges. Instagram's leadership has explicitly emphasized a pivot toward messaging and intimate exchanges, reflected by interface changes that prioritize DMs, Reels, and short-form exchanges over legacy photo feeds (Campbell, 2025). Threads has likewise rolled out direct and group messaging as a deliberate move away from a purely public, feed-based model (Meta, 2025; Perez, 2025). Industry analysis echoes this broader transition: Users increasingly shift engagement into private channels, with TikTok seeing rapid growth in DM-based sharing and declines in public commenting (Harmon, 2025). Platforms such as Discord exemplify this interactional turn, organizing participation around server-based conversations rather than mass visibility (Discord, 2025; Patel, 2024). These developments underscore that chatroom-like interaction environments are not peripheral to social media but increasingly central to it. Yet empirical research continues to rely disproportionately on public, feed-based contexts, leaving open questions about how core interactive features operate in conversational spaces where participation thresholds, visibility dynamics, and power concentration may differ fundamentally.

Our work is grounded in the Theory of Interactive Media Effects (TIME) (Sundar et al., 2015), which models how interactivity shapes user perceptions and behavior; yet the theory has not distinguished between the human–system and human–human behaviors made possible through interactivity (McMillan, 2010; Rafaeli, 1988). Furthermore, while prior work has treated interactivity as a process, function, or perception, few studies examine its consequences for emergent behaviors on platforms. For instance, reaction tools (e.g., upvotes, likes) and response mechanisms (e.g., replies, mentions) are intended to lower barriers and promote exchange (Centola et al., 2018; Esau et al., 2017; Poor, 2005), yet their effects may diverge for different discussion goals or marginalized communities (Auxier & Anderson, 2021; Matias, 2019; Postmes et al., 2001; Stromer-Galley, 2007).

This study focuses on the interactive features of social media platforms that allow users to react to content or reply to others, and offers three contributions. First, we have developed and will release a custom-built social media platform that allows experimental manipulation of reaction and reply features. Second, we introduce and operationalize the concept of internal inclusiveness: who contributes, how they engage, and which perspectives gain visibility. Third, our experiments provide causal evidence on how different forms of interactivity shape discussion dynamics: Reactions reduce overall engagement but increase exposure to diverse viewpoints by lowering participation thresholds, while replies foster deeper exchanges but concentrate discussion power among select users. Viewed as a whole, the study advances theoretical understanding of feature-level trade-offs in online discussion and helps identify design choices that foster deeper engagement without compromising equitable participation.

Background

Political discussions in the public sphere ideally support inclusive deliberation, where participants engage actively, exchange perspectives, and consider diverse viewpoints before forming opinions (Gastil, 2008; Habermas, 1984). On social media, however, inclusiveness is often constrained by participation asymmetries: A small subset of users produces most content while the majority engages passively. For example, while Twitter/X reported 398 million monthly active users in 2022, only 10% tweeted on any given day, underscoring the gap between access and active participation (Pfeffer et al., 2023). These disparities highlight that open access alone does not ensure inclusiveness: Platform design shapes who participates, how interaction unfolds, and which perspectives become visible.

Interactivity has been conceptualized in distinct but complementary ways. Rafaeli (1988) emphasized contingency in message exchange, Stromer-Galley (2004) distinguished interaction-as-process from interaction-as-product, and McMillan (2010) differentiated human–human, human–document, and human–system interaction. The TIME model shifted the focus from taxonomy to mechanism by specifying how concrete interface features shape users' psychological experience (Sundar et al., 2015). Collectively, these frameworks demonstrate that interactivity is multifaceted, encompassing the properties of a system, the experience of agency, and the communicative exchanges that emerge between users.

What these frameworks do not specify, however, is which concrete, manipulable design features produce which downstream behavioral outcomes. Because our study randomizes specific interface features (reactions and replies) we adopt a feature-level operationalization. We treat a design element as interactive to the extent that it affords reciprocal responsiveness between users, focusing on the interpersonal behaviors each feature makes possible. The manipulated variable is the human–system action (reacting or replying), and the theorized consequences occur in the human–human domain (participation, engagement, and diversity).

This distinction also clarifies our use of "low-effort" and "high-effort" features. We use these terms in line with established typologies that distinguish interface actions by the activity and cognitive involvement they require. Flores and Hilbert (2023) differentiated lean-back, low-effort behaviors such as reactions from lean-forward, high-effort behaviors such as replies. Wang and Sundar (2022) similarly separated cue-based, low-activity interactions from action-based, high-activity interactions—a distinction that maps directly onto the contrast between clicking a reaction icon and composing a textual response. Ellison et al. (2020) further situated viewing, clicking, and commenting along a gradient of user effort. Our terminology, therefore, reflects the consensus in the literature on differences in the activity and cognitive involvement required for specific interface actions.

Because low- and high-effort actions change who can participate and what kinds of contributions become visible, they have direct implications for what we call *internal inclusiveness—participatory equality within a discussion*. This focus highlights feature-level trade-offs: design elements may lower the barrier for visible participation while simultaneously narrowing the distribution of substantive contributions. It also underscores how system-level actions shape not only individual perceptions but also the broader distribution of voices in political conversation.

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Important puzzles remain about how different features shape participation, engagement, and diversity—and, in turn, the internal inclusiveness of political discussion. Low-effort features such as reactions may lower the threshold for entering a discussion yet alter the substance and distribution of contributions. High-effort features such as replies may deepen interpersonal engagement while discouraging contributions from less experienced or dissenting users. To examine the causal effects of these features, we employ a custom-built platform that allows experimental manipulation of human–system actions (reacting, replying) and observation of the human–human interaction patterns that follow.

Hypotheses and research questions

Our experiment isolates two platform features—reactions and replies—that differ in the level of effort required from users and in the forms of interpersonal exchange they enable. These features operate at the level of human–system interaction: they determine whether a user can respond through a single click or must produce a directed textual message. Our theoretical interest, however, lies in the downstream human–human interaction that these system-level features shape. We measure three behavioral outcomes: participation (who contributes, how often, and how much), engagement (the degree of reciprocal, directed interaction among users), and diversity (the range of viewpoints expressed). In short, we manipulate human–system actions and observe the human–human processes they influence.

The reaction feature

Reactions lower the threshold for visible contribution by enabling a one-click signal (Flores & Hilbert, 2023). Prior work suggests that such lightweight interactions can meaningfully shape subsequent behavior: They alter attention flows and influence which contributions receive visibility (Bakshy et al., 2011; Centola, 2019; Vosoughi et al., 2018). At the same time, they may reduce substantive interpersonal exchange because they function as conversational endpoints (Levordashka et al., 2021; Meier et al., 2014), create visibility bottlenecks that privilege a small subset of posts (Guess et al., 2023), or satisfy acknowledgment needs without prompting further interaction (Wang & Sundar, 2022). Together, these mechanisms imply that enabling a low-effort reaction feature should reduce both participation (who contributes, how often, and how much) and interpersonal engagement.

H1: Relative to conditions where reactions are unavailable, the availability of message-specific reactions reduces (a) user participation and (b) interpersonal engagement in discussions.

The implications of reactions for diversity are theoretically mixed. Reactions may narrow diversity by amplifying majority viewpoints (Bakshy et al., 2011; Guess et al., 2023) or broaden it by elevating scattered signals of cross-cutting content (Guess & Coppock, 2020; Wuestenenk et al., 2025). Because the direction is ambiguous, we pose a research question:

RQ1: How do message-specific reactions affect the diversity of political discussions?

The reply feature

Replies constitute a higher-effort form of human–system interaction: users must generate text, address others directly, and sustain a conversational thread. Evidence suggests that reply pathways increase the likelihood of reciprocal exchange (Lew & Stohl, 2023; Velasquez, 2012), extend conversation length through deeper threads (Adelani et al., 2020; Lin et al., 2021), and facilitate relational connection (Granovetter, 1973; Pan et al., 2017). These mechanisms imply that enabling reply functionality should increase participation and interpersonal engagement.

H2: Relative to conditions where replies are unavailable, the availability of message-specific replies increases (a) user participation and (b) interpersonal engagement in discussions.

The consequences of replies for diversity are likewise mixed. Replies may deepen topical exploration (Bakshy et al., 2011; Centola, 2019; Lee et al., 2018), but they may also produce siloed conversational clusters centered on a small number of dominant posts (Lin et al., 2014). To adjudicate between these mechanisms, we ask:

RQ2: How do message-specific replies affect the diversity of political discussions?

Method

We conducted a 2-by-2 between-subjects factorial experiment to examine how two common platform features—reactions and replies—shape communication patterns in synchronous political discussions. The manipulation operated entirely at the level of human–system interaction: Participants were given or denied the ability to (a) register a reaction to a message using upvote/downvote buttons, and/or (b) post directed textual replies that created explicit conversational threads. Our analysis focuses on the downstream human–human interaction these features produce: who contributes, how they respond to others, and the diversity of perspectives expressed.

To implement this design, we custom-built a browser-based platform in JavaScript, modeled after real-time, chat-based environments such as WhatsApp or Discord. These conversational platforms represent a broader class of contemporary social media spaces where political talk increasingly occurs, and where reacting and replying are central interaction mechanisms. Participants were automatically assigned to chatrooms, with feature conditions randomized at the room level. Each room was assigned one of four feature sets: both reaction and reply enabled, reaction enabled only, reply enabled only, or neither

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enabled. Features were always tied to individual messages rather than applied globally. For example, when a participant sent a message, a set of glyphs (upvote, downvote, reply) appeared alongside it only if those features were active in that condition. Our Supplementary Materials provide further technical details and screenshots of the interface.

Recruitment

Participants were recruited from CloudResearch's MTurk panel over a 4-month period during which debates over social media content regulation were salient in U.S. public discourse. Eligibility criteria required participants to (a) reside in the United States, (b) be at least 18 years old, (c) use a laptop or desktop computer, and (d) have an approval rate above 80%. CloudResearch quality controls (e.g., geolocation verification, duplicate IP blocking) were applied to preserve data integrity.

The HIT advertisement stated that participants would join a *30-minute live policy discussion* for a $6 bonus, in addition to $0.50 for a short pre-survey. Participants who joined the waiting room but could not be matched were paid $1. All amounts exceeded the federal minimum wage for the expected time commitment. HITs were scheduled across multiple weekday sessions to allow real-time matching into groups of five.

A power analysis using the pwr package in R, with an expected effect size (Cohen's d) of 0.2, $\alpha = 0.05$, and power = 0.80, indicated that 393 participants per group (1,572 total) would be required. In total, $N = 1,643$ participants completed the study.

Procedure

After completing a pre-survey on demographics and policy attitudes, participants entered a virtual waiting room. Once five participants were present, they were placed into a new chatroom that had been randomly assigned one of the four feature conditions.

Before the discussion began, participants viewed a 1-min instruction page describing the features available in their assigned condition. The page remained accessible throughout the session via an "Instructions" button. A pinned message restated the topic and encouraged use of the available features. For instance, in the reply-only condition, the pinned message encouraged participants to direct their messages as replies. We added this brief prompt to approximate the minimal guidance platforms often provide when introducing features (e.g., sidebar cues on Reddit or onboarding tips in Discord).

Each discussion lasted approximately 28 min. Five system-generated prompts were posted at fixed intervals to sustain conversation, each addressing a different dimension of the policy topic (e.g., risks/benefits of regulation, free speech, platform governance). The interface displayed all messages chronologically in a shared chat window, with reaction/reply buttons appearing next to each message depending on the condition. No human moderators were present, and off-topic deviations were rare.

After the discussion, participants completed a post-survey assessing perceived discussion quality and post-treatment attitudes. Those who completed all components received the full $6 bonus. Figure 1A summarizes the design, and Table 1 reports sample sizes by condition.

Dependent variables

Prior work demonstrates that interactive features shape different aspects of participation and discussion quality, including how people contribute, how they interact, and which perspectives gain visibility (Auxier & Anderson, 2021; Centola et al., 2018; Esau et al., 2017; Matias, 2019; Postmes et al., 2001; Stromer-Galley, 2007). Building on deliberative scholarship (Gastil, 2008; Habermas, 1984; Steenbergen et al., 2003) and network approaches to online discussion (Flores & Hilbert, 2023), we conceptualize *internal inclusiveness* along three behavioral dimensions:

- Participation: who contributes, how often, and how much.
- Engagement: how users interact with one another through directed exchanges and conversational structure.

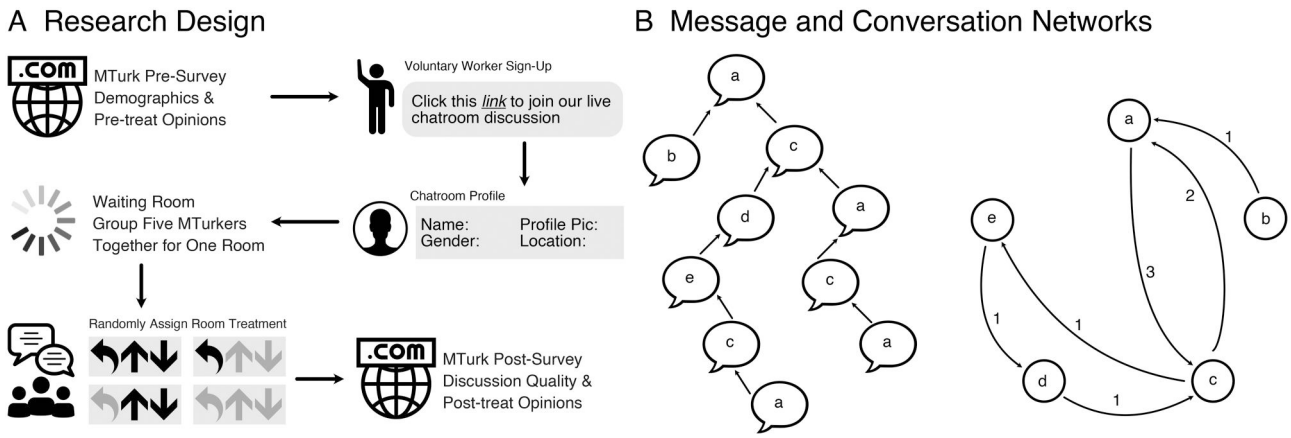


Figure 1 Research design and network construction. (A) Research design. (B) Message and conversation networks. *Note*: The network on the left-hand side of Panel B shows the message network, where nodes are participants' messages, and ties are reply logic. The network on the right-hand side of Panel B shows the conversation network, where nodes are participants and ties are weighted to indicate the times a participant replies to another participant.

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Table 1 The number of subjects per treatment condition ($N = 1{,}643$).

Treatment conditions			
		Allow reactions	
		No	Yes
Allow Replies	No	399	408
	Yes	420	416

- Diversity: the range of viewpoints, topics, and structural pathways represented in the discussion.

Each dimension is measured at two levels of analysis. User-level measures capture individual behavioral profiles; room-level measures, defined as the median of user-level scores within each chatroom, capture collective discussion patterns. Reporting both levels allows us to distinguish individual behavioral shifts from group-level dynamics. These dimensions draw on direct usage traces and network-based measures, reflecting how individual actions aggregate into collective patterns of interaction and exposure.

Participation measures

Participation concerns the basic behavioral footprint of a user's contributions, captured concretely as who contributes, how frequently they do so, and how much they contribute.

Following online discussion research, contribution elaboration is operationalized using observable message characteristics such as length, which provides a behavioral indicator of how much content a user contributes (Claridge, 2018; Jaidka et al., 2019).

We measure:

- Message volume: the total number of messages a user sends.
- Message elaboration: median message length (in words), used as a behavioral measure of contribution substantiveness.
- Contribution scope: number of distinct users a participant replies to or addresses (explicitly or implicitly), corresponding to out-degree in the reply network.

Engagement measures

Engagement captures how participants relate to one another through reply-based or directed interactions. Network-analytic approaches conceptualize engagement structurally, using the depth and cohesion of reply chains to indicate reciprocal exchange (Choi et al., 2011; Pan et al., 2017, 2020). We follow this tradition by constructing a directed reply network in which edges represent functional replies (either formal "Reply" clicks or implicit conversational references via human annotation). From this network, we compute:

- Closeness centrality: how efficiently a user can reach others through conversational paths.
- Triadic closure: the extent to which a user's conversational partners also interact with one another.
- Conversation depth: the maximum nesting level of reply chains, that is, how many successive back-and-forth exchanges occur before a thread terminates. This vertical measure of thread structure is distinct from breadth (the number of parallel branches) and captures sustained dyadic or small-group exchange rather than topical spread.
- Density (room level): the proportion of realized participant-to-participant ties in the room.

Diversity measures

Diversity reflects variation in perspectives, topics, and structural exposure. Following research on structural diversity, entropy-based measures, and cross-cutting information flow (Ahmed et al., 2016; Choi et al., 2011; Gallagher et al., 2021; Lin et al., 2014; Shen & Cage, 2015; Vosoughi et al., 2018), we examine three dimensions.

- Structural diversity:
  - Betweenness centrality: whether a user connects otherwise segregated discussion clusters.
  - Structural virality (room level): whether discussion spreads through long conversational chains rather than star-shaped patterns.
- Viewpoint diversity:
  - Shannon diversity: entropy of stance proportions for each user or room, capturing ideological variation.
- Topic diversity:
  - Topic entropy: breadth of topics discussed by a user or across a room.
  - KL divergence: divergence between an individual's topic distribution and the room distribution, indicating the distinctiveness of their contributions.

Deriving content features

In the reply condition, participants could attach comments directly to specific messages in a threaded format. Because the presence of a reply feature may also encourage implicit responses (e.g., addressing prior points textually), we treated all responsive contributions as replies for participation and engagement analyses. This captures both the mechanical effect of the feature and the conversational norm it establishes.

To identify functional replies and stance positions, we conducted a human-annotation task. Full details are provided in the Supplementary Materials, including inter-annotator agreement statistics. Briefly, reply annotations yielded 70.64% pairwise agreement and a Cohen's Kappa of 0.55; stance annotations achieved 88.26% agreement with a Kappa of 0.41. These values fall within accepted ranges for multiclass, subjective coding tasks (Artstein & Poesio, 2008; Ng & Carley, 2022). Following recent recommendations to supplement chance-corrected metrics (Barriere et al., 2023; Davani et al., 2022; Jaidka et al., 2024), we also computed $\mathit{0}$, which accounts for label prevalence using expectation maximization (Passonneau & Carpenter, 2014). Reliability exceeded the recommended threshold of $\mathit{0} \geq 0.65$, with scores of 0.70 (reply) and 0.77 (stance).

Topic-level features were derived using BERTopic (Grootendorst, 2022). Message embeddings were clustered

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through coherence-guided model selection, resulting in a 20-topic solution. Each message was represented as a probability distribution over these topics, enabling the computation of entropy- and divergence-based diversity measures. As reported in the Supplementary Materials, the resulting themes primarily reflected (a) stakeholders (companies, advertisers), (b) regulation, (c) speech-related debates, and (d) conversational fillers. Table 2 presents a sample of annotated conversations.

Analytical strategies

We regressed our dependent variables on two interactive feature treatments and their interaction term, controlling for user demographics, that is, age ($M = 39.16$, $SD = 12.52$), sex (Female = 49.12%),[1] and partisan identity (Strong Democrat = 26.6%, Weak Democrat = 32.1%, Independent = 10.7%, Weak Republican = 19.1%, Strong Republican = 11.3%).

A minimum of 399 and a maximum of 416 participants were assigned to each of the four treatment conditions. Treatments were entered as binary predictors in the generalized linear models. Models were estimated using the *lme4* package in R, with random intercepts accounting for the non-independence of observations within chatrooms. All measures were min–max rescaled to the [0,1] range at the user level. Standardized effect sizes were computed using the *effectsize* package, based on F values from the linear models. Analyses were conducted at both the user and room levels: room-level measures were defined as the median of user-level scores within each chat session. All regression results were Bonferroni-adjusted for multiple comparisons.

Results

Recruitment

In total, 4,236 MTurkers took our pre-survey ($0.5). Of these, 2,335 clicked our invitation link and joined the waiting room, showing an intention to participate in the study. After waiting 10 min without matching with other participants, 724 were thanked and partially compensated ($1.5). Ultimately, 1,643 users joined our main study and were assigned to 302 rooms with random combinations of reply and reaction buttons ($6.5). Among them, 273 users did not send any messages.

Thereafter, 1,370 participants sent 25,016 messages across these rooms, averaging 81 per chat. Three thousand eighty-two messages were replies to other messages, which 473 users created. Furthermore, 6,785 messages received 10,603 reactions (9,484 upvotes and 1,119 downvotes) from 688 users. The average message length was 89 characters (median was 77 characters), and the maximum message length was 301 characters. Message-level scores were aggregated to the user level for subsequent analyses.

[1] The full sex distribution was Female = 673, Male = 689, and Other = 8. Due to the small number of cases in the "Other" category, we categorized sex as Non-Male vs. Male for analysis.

The effect of reactions

As seen in Figure 2, **H1a was partially supported**, suggesting that low-effort reactions reduced the breadth of participation but encouraged slightly more elaboration in each message. The ability to react decreased the number of user messages ($\beta = -0.053$, $SE = 0.009$, $p < .001$) and their contribution scope ($\beta = -0.079$, $SE = 0.022$, $p < .001$). It also increased message length ($\beta = 0.042$, $SE = 0.011$, $p < .001$). These effects were consistent at the room level: fewer messages ($\beta = -0.053$, $SE = 0.010$, $p < .001$), longer messages ($\beta = 0.042$, $SE = 0.011$, $p < .001$), and lower contribution scope ($\beta = -0.077$, $SE = 0.022$, $p < .001$).

As seen in Figure 3, **H1b was partially supported**, suggesting that low-effort reactions weakened the cohesion of interaction networks without deepening conversations. Reactions reduced several indicators of engagement. Users showed lower closeness centrality ($\beta = -0.077$, $SE = 0.022$, $p < .001$) and formed fewer triadic closures ($\beta = -0.061$, $SE = 0.021$, $p < .01$). Networks also became less dense at the room level due to the reaction button ($\beta = -0.155$, $SE = 0.041$, $p < .001$). In contrast, reactions had no effect on conversation depth ($\beta = 0.001$, $SE = 0.005$, $p > .10$). These findings were consistent at the room level (closeness, $\beta = -0.080$, $SE = 0.021$, $p < .001$; triadic closure, $\beta = -0.060$, $SE = 0.021$, $p < .01$).[2]

In answer to RQ1, Figure 4 shows that the ability to react predicted higher diversity across three of six measures. It increased the betweenness ($\beta = 0.060$, $SE = 0.020$, $p < .01$) and Shannon diversity ($\beta = 0.055$, $SE = 0.019$, $p < .01$) at both the user and room levels. The structural virality also increased at the room level ($\beta = 0.127$, $SE = 0.058$, $p < .05$). On the other hand, the ability to react decreased the topic entropy at both levels (user-level, $\beta = -0.035$, $SE = 0.011$, $p < .01$; room-level, $\beta = -0.034$, $SE = 0.011$, $p < .05$), suggesting that people were more likely to stick with popular topics (i.e., herd behavior), likely aiming to increase the chances of their messages being upvoted.

The effect of replies

As seen in Figures 2 and 3, **H2a was not supported** and **H2b was partially supported**. The reply feature increased conversation depth at both the user level ($\beta = 0.022$, $SE = 0.007$, $p < .01$) and the room level ($\beta = 0.022$, $SE = 0.005$, $p < .001$). This suggests that replies helped users sustain longer discussion chains even though they did not strengthen other forms of engagement. However, the reply feature did not improve users' closeness ($\beta = 0.030$, $SE = 0.022$, $p > .1$), triadic closure ($\beta = 0.006$, $SE = 0.021$, $p > .1$), or network density ($\beta = 0.009$, $SE = 0.041$, $p > .1$).

In response to RQ2, Figure 4 shows that the reply feature had mixed effects on discussion diversity. Its most consistent impact was a reduction in viewpoint diversity: the ability to reply decreased the Shannon diversity of user stances ($\beta = -0.060$, $SE = 0.019$, $p < .01$), an effect that held at both the user and room lvevel. Conversely, the feature showed a small positive effect on topic diversity, increasing the KL Divergence of messages

[2] The post-hoc power analysis, conducted using the simr package for our mixed-effects model, revealed a substantial power of 90% to detect the effect of the "Allow Reactions" treatment. This indicates a strong likelihood that our study was adequately powered to identify significant effects associated with this predictor.

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Table 2 Excerpts of chats from rooms implementing different treatment conditions.

Allow reactions	Allow replies	UserId	Sample messages	Upvotes	Downvotes	Reply to	Room Id	MessageId	Automatically labeled topic
		6	Well, obviously free speech comes into any discussion about regulation.	3	0	0	2	4	Free speech
		5	I think censorship is not only good but necessary for removing provably false and harmful (i.e., bullying, slander, etc.) speech. But it shouldn't be used for political silencing.	3	0	0	2	5	Free speech
Yes	No	6	Totally agree.	1	0	0	2	6	Agreement
		7	I agree but feel it is better for companies to moderate than the government.	2	0	0	2	10	Companies
		5	I don't care if it's removing Alex Jones or Chapo House, left or right, they're just political commentators. The Government especially shouldn't be able to censor political speech if it's not demonstrably false or harmful.	3	0	0	2	14	Free speech
		4	I just wish companies would admit to their bias and political leaning.	3	0	0	2	15	Companies
		5	Agreed, no social media is completely secular.	3	0	0	2	22	Social media
		7	On Twitter there are lots of people who aren't real people but fake profiles working for corporations and enemy governments. I don't think anonymous people should have free speech.	2	0	0	2	24	Free speech
		5	We're anonymous lol	3	0	0	2	26	Conversation fillers
		121	I think advertising should be subjected to certain regulations too.	0	0	0	19	1297	Advertising
		123	Agree	0	0	1297	19	1299	Agreement
		120	TV ads are regulated, why shouldn't social media ads be?	0	0	0	19	1301	Advertising
		121	You should not be able to publish things that will incite violence or terrorism.	0	0	0	19	1303	Organization
No	Yes	122	No I don't think so because its pretty evident some political advertising is just straight up hate speech	0	0	0	19	1306	Advertising
		120	Agreed	0	0	1306	19	1312	Agreement
		124	As far as advertising, they can advertise what they want. If their advertisements cross lines or in bad taste their bottom line could hurt as people will leave that platform.	0	0	0	19	1316	Advertising
		120	It could have even worse consequences than that … inciting violence, riots, etc.	0	0	0	19	1326	Free speech

(continued)

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Table 2 (continued)

Allow reactions	Allow replies	UserId	Sample messages	Upvotes	Downvotes	Reply to	Room Id	MessageId	Automatically labeled topic
		121	Not all political speech is hate speech. However, it should not be used to allow them to spread straight lies.	0	0	0	19	1331	Hate speech
		123	But you can see that it hasn't. Advertising needs to be regulated.	0	0	1316	19	1333	Advertising
		120	Look at Parler (Parlor? not sure how to spell it)	0	0	0	19	1343	Companies
		68	I do feel people should have free choice in their statements, but it should be regulated. If it causes harm to others, then it should not be said	0	0	0	10	583	Free speech
		67	I think that as long as what they are saying does not encourage or promote harm to others they should be allowed to say whatever they want.	0	0	0	10	589	Hate speech
No	No	67	I also think that is a very important question that was mentioned above, WHO decides what is right or wrong, I would say there should be some sort of vote by the users themselves and not outside forces	0	0	0	10	599	Platforms
		69	Most SM platforms allow the users to mute offensive content, so it's not necessary for the platforms to do it.	0	0	0	10	601	Platforms
		69	The word "harm" has no meaning anymore.	0	0	0	10	611	Hate speech
		68	It could be whomever. It could be the government or a corporation. It doesn't take my freedoms away. I am not a person who writes incitement and harmful things.	0	0	0	10	619	Overseers
		73	Well, in answering the question we're provided, it seems obvious we're not free to say whatever we want, given that doing so results in getting banned and perhaps canceled.	0	0	0	10	639	Free speech
		68	I feel that the social media platforms should do the governing. That way, if a person doesn't like the amount of speech limited they can choose another platform	0	0	0	10	645	Platforms
		67	I dont think it matters referring to the question is social media companies govern or the government itself, it would have the same result regardless	0	0	0	10	650	Advertising

(continued)

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Table 2 (continued)

Allow reactions	Allow replies	UserId	Sample messages	Upvotes	Downvotes	Reply to	Room Id	MessageId	Automatically labeled topic
		67	and if you think about it if the social media companies regulate in a way the government does not like then they can force the social media companies to do what they want anyway, and somewhat vice versa	0	0	0	10	668	Advertising
		70	I think social media already too much power because of certain regulations that protect them	1	0	0	11	730	Overseers
		71	Yes, and they're already violating the rules that protect them to have even more power.	0	0	0	11	734	Agreement
Yes	Yes	75	It should decrease slightly or stay the same.	0	0	0	111	735	Trends
		70	I think content regulation should be relaxed heavily, and enforced in clear cut cases of crime and violence	0	0	0	11	736	Organization
		74	I agree with decreasing slightly.	1	0	0	11	738	Trends
		72	Content regulation should be increased. The last year should be proof of that.	0	3	0	11	739	Trends
		71	It should decrease significantly. That is how they agreed to operate initially. It's what protects them from liability for the content their users post.	1	1	0	11	743	Trends
		71	You live in bizzarro world.	0	1	739	11	747	Greetings
		72	Oversight is needed	1	1	0	11	749	Overseers
		71	They've already proven that their idea of oversight is censorship. Also, again, they're already violating the agreements they made in how they were going to operate.	0	1	0	11	756	Companies
		72	Regulation should not censor free speech but help keep that speech truthful.	1	1	0	11	760	Free speech
		71	Untrue speech (other than libel and slander) is protected by the constitution as much as true speech.	0	1	0	11	764	Free speech
		74	It would be untrue of me to say I agree	0	0	764	11	766	Agreement

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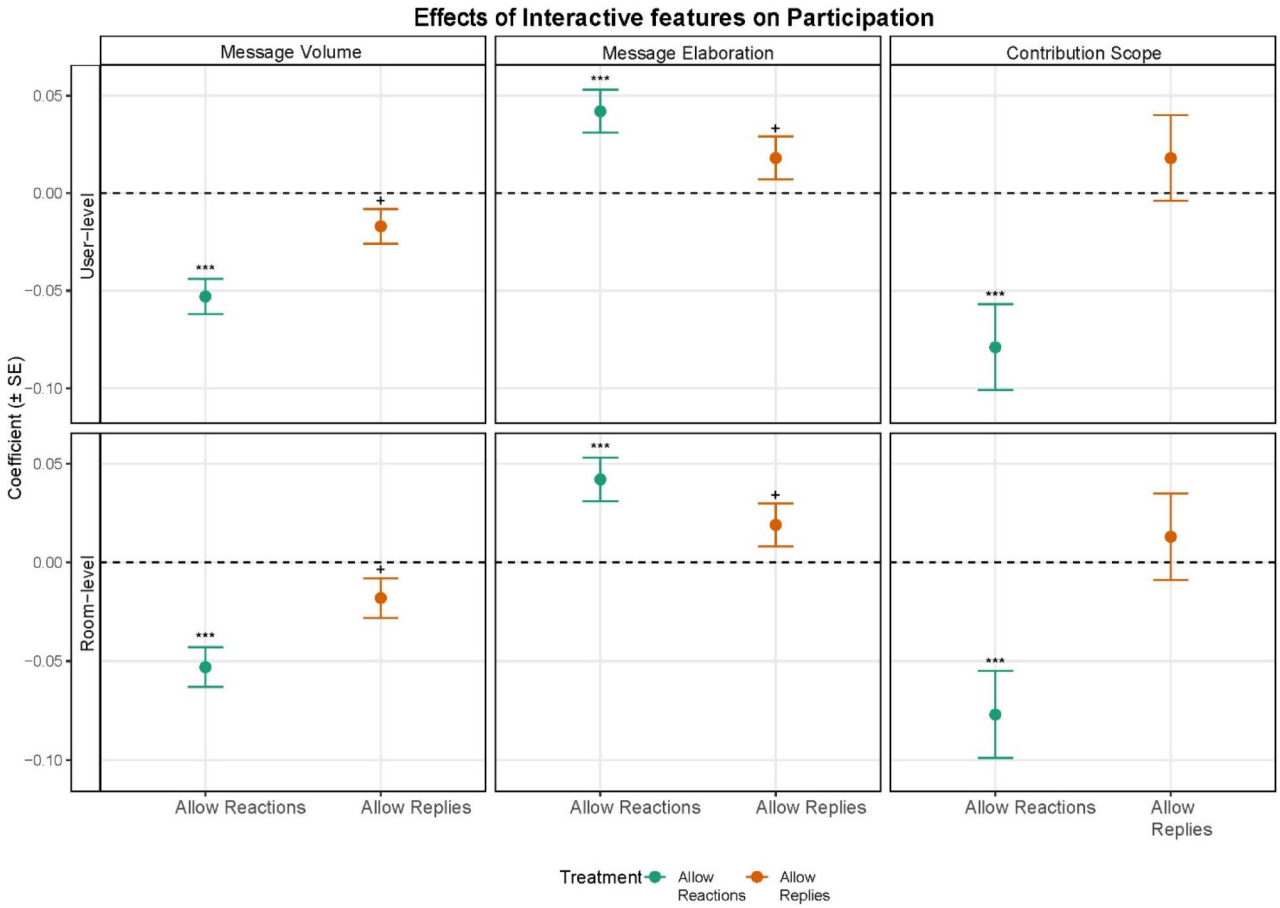


Figure 2 The effects of reaction and reply on measures of participation. *Note*: + $p < .1$, * $p < .05$, ** $p < .01$, *** $p < .001$.

at the user level ($\beta = 0.014$, $SE = 0.003$, $p < .001$), though this effect was not significant at the room level. This indicates that while replies narrowed viewpoint expression, they may have encouraged users to introduce new perspectives within a topic.

Discussion

This study examines how system-level design features—whether an interface allows users to react with a single click or to produce a directed textual reply—shape the interpersonal dynamics of political discussion. Specifically, we manipulate forms of human–system interaction and assess their downstream consequences for human–human interaction, captured through participation, engagement, and diversity. Taken together, the central take-home is that no single interactive feature uniformly enhances discussion quality: Reactions trade participation breadth for viewpoint diversity, while replies trade viewpoint diversity for conversational depth. Three key findings emerge.

First, replies deepen conversational exchange without expanding the overall interaction network, whereas reactions reduce the frequency of contributions even as they increase the elaboration of the messages users choose to produce. Second, the results show that internal inclusiveness is inherently multidimensional: Interface features produce opposing effects across participation, engagement, and multiple forms of diversity, revealing distinct behavioral pathways that matter for different communication goals. Third, the study offers a generalizable experimental framework for feature-level evaluation, enabling causal tests that are rarely possible in platform-sanctioned settings.

Prior work often treats interactivity as a unitary construct, thereby obscuring how specific features can produce distinct expressive, relational, and structural patterns in discussion (Sundar et al., 2015). Our experiment isolates reactions and replies and traces their behavioral consequences, extending observational research on how these structures organize discussion.

Table 3 summarizes our findings and reveals a central insight: Features can shift participation, engagement, and diversity in different directions, consistent with deliberative accounts that distinguish these components of discussion (Bächtiger, 2019; Gastil, 2008; Steenbergen et al., 2003).

The results for participation and engagement make this clear. Replies foster greater conversational depth while leaving breadth unchanged, illustrating how a feature can amplify specific forms of engagement without expanding the overall interaction structure. This pattern is consistent with evidence that reply features support sustained, dyadic or small-group exchanges rather than uniformly expanding interaction networks (Lin et al., 2021; Pan et al., 2017, 2020; Velasquez, 2012). Reactions, in contrast, lowered baseline participation by reducing message volume and contribution scope, yet simultaneously

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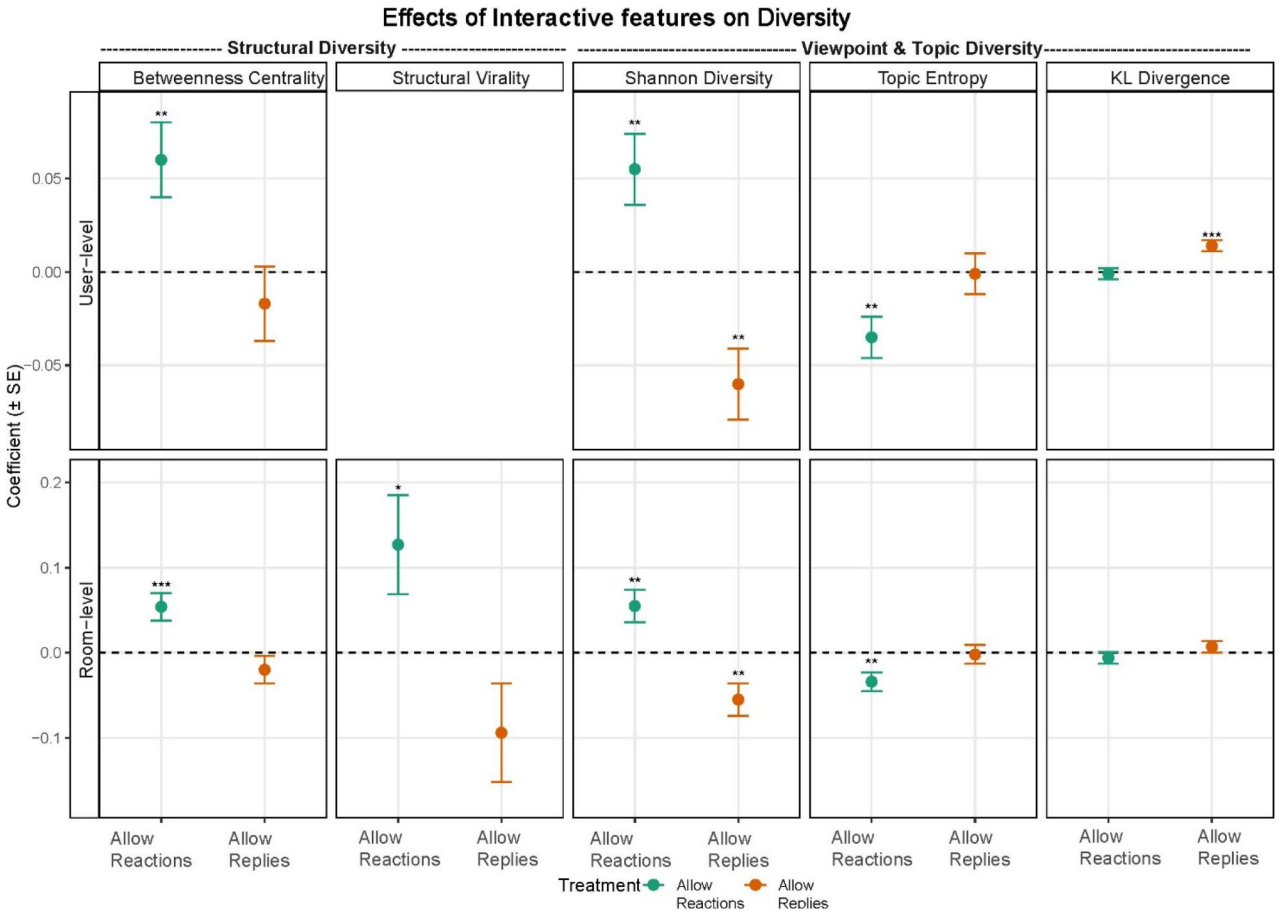


Figure 3 The effects of reaction and reply on measures of engagement. *Note*: + $p < .1$, * $p < .05$, ** $p < .01$, *** $p < .001$.

increased message elaboration. Users relied on reactions as a behavioral substitute for posting, contributing less frequently but investing more effort when they did participate. This combination aligns with work suggesting that low-effort signals can act as conversation terminators (Levordashka et al., 2021) or satisfy acknowledgment needs (Wang & Sundar, 2022), while also encouraging more deliberate contributions from those who choose to participate (Claridge, 2018; Papakyriakopoulos et al., 2023).

The diversity results extend this logic by showing how features channel attention into different behavioral pathways. Replies anchored users within conversational threads that elicited new thematic elaborations but did not broaden ideological range. This echoes prior findings that reply chains can deepen topical focus while encouraging clustering around particular positions (Lee et al., 2018; Lin et al., 2014). Reactions, meanwhile, widened exposure to distinct stances while concentrating attention around a narrower set of topics. This pattern parallels evidence that endorsement-based mechanisms can amplify salient posts and stances (Bakshy et al., 2011; Guess et al., 2023), even as they shape which viewpoints are surfaced and circulated (Guess & Coppock, 2020; Vosoughi et al., 2018). Together, these results demonstrate that diversity is not a single construct: Structural, topical, and ideological diversity respond differently to design choices, consistent with entropy-based approaches to measuring informational diversity in online settings (Ahmed et al., 2016; Gallagher et al., 2021; Shen & Cage, 2015).

These dynamics complicate assumptions that low-effort features uniformly erode discussion quality or that high-effort features uniformly enhance it. They also have broader implications for the design and evaluation of interactive systems. Studies of online discussion often rely on a single indicator—typically viewpoint diversity or message volume—which risks obscuring how features shape political discourse through divergent behavioral pathways (Flores & Hilbert, 2023; Hopp & Vargo, 2019; Kim et al., 2021). The present findings show that features supporting contextual elaboration do not necessarily broaden ideological exposure, just as features that increase structural reach may compress topical breadth. Consequently, no single measure captures the full behavioral impact of an interactive design choice. Experimental evaluations benefit from multidimensional operationalizations that reflect the varied forms of inclusiveness relevant to different communication and policy goals (Fung, 2006; Han et al., 2015; Karpowitz & Raphael, 2016).

Overall, the study extends the TIME model by demonstrating that interactive features shape inclusiveness across multiple, sometimes opposing, behavioral dimensions (Sundar et al., 2015; Wang & Sundar, 2022). The results show why designers cannot assume that increasing interactivity uniformly enhances discussion quality. Instead, interactive features must be aligned with the particular forms of inclusiveness a platform seeks to

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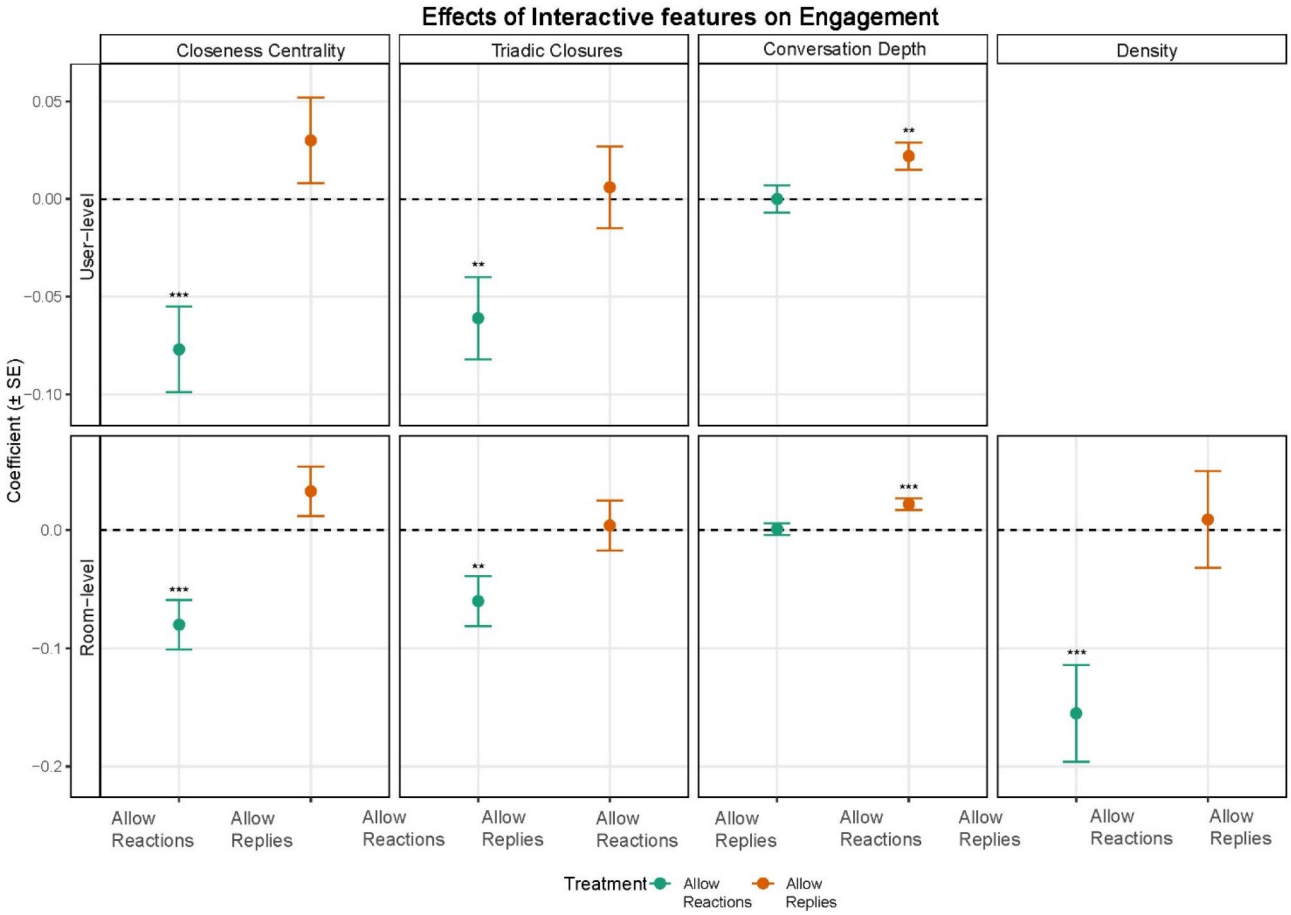


Figure 4 The effects of reaction and reply on measures of diversity. *Note*: + $p < .1$, * $p < .05$, ** $p < .01$, *** $p < .001$.

promote—whether richer contextual elaboration, sustained cross-cutting exposure, or more equitable participation. The findings illuminate how specific design trade-offs affect both individual perceptions and the distribution of voices and perspectives in online political conversations, especially as platforms shift toward chat- and DM-based environments.

Limitations

While the present study provides a foundation, the interplay of social norms, the contentiousness of issues, and the rate of norm violations may vary across national and cultural contexts. We focused on an American cohort deliberating a social media policy issue in Congress, where partisan identities are highly accessible, moralized disagreement is common, and platform talk is often framed through free-speech norms. In such settings, reactions may be especially prone to being used for signaling, pile-ons, or strategic endorsement, and reply features may disproportionately reward confident or partisan-leaning contributors. Other behaviors might arise in other contexts with different norms.

While our study emphasizes inclusive political discussions, inclusiveness does not inherently guarantee civility or high deliberative quality. Deliberative features can create trade-offs for civility and constructiveness (Jaidka, 2022). Future research should examine how participation, engagement, and diversity align—or conflict—with standards such as civility and reason-giving.

Our interface resembled platforms such as Twitter, Reddit, or Discord, where anonymity and one-off interactions are common. We conceptualized "social media" as a bundle of affordances—design elements that enable or constrain user behavior (Bucher & Helmond, 2018). Our scope conditions thus apply to environments with synchronous or near-synchronous text exchange, minimal identity verification, and lightweight interactive features. We prioritized psychological realism (i.e., activating the same cognitive and emotional processes as real-world interactions) over mundane realism (i.e., surface resemblance) (Druckman, 2022). The goal was not to reproduce every sociocultural nuance of real platforms but to preserve the affordances most relevant to our causal questions: reactions, replies, and real-time engagement. However, we recognize that a few exogenous design choices, such as the brief instructional prompts, may have increased the salience of these features more than would occur in entirely naturalistic settings.

Our experiment simulated a single, prompt-driven conversation among Amazon Mechanical Turk (AMT) workers. While AMT samples are widely used and yield valid data (Berinsky et al., 2012), their motivations differ from casual social media users, who often engage for social connection or identity expression. These differences may limit generalizability. A further limitation

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Table 3 How reaction and reply features shape different dimensions of participation, engagement, and diversity.

Dimension	Measure	What it captures	Reaction effect	Reply effect	Interpretive note
Participation					
Message Volume	Total messages sent	Baseline activity	↓	(marginal) ↓	Reactions substitute for posting; replies minimally affect volume
Message Elaboration	Median message length	Cognitive effort per post	↑	(marginal) ↑	Fewer messages but more effort per contribution
Contribution Scope	Number of distinct users addressed	Breadth of participation	↓	null	Reactions narrow participation networks; replies do not broaden them
Engagement					
Closeness Centrality	Avg. distance to others	Connectivity in network	↓	null	Reactions suppress network cohesion; replies affect other dimensions
Triadic Closure	Closed triangles among users	Local clustering	↓	null	Reactions reduce mutual engagement; replies do not increase clustering
Conversation Depth	Length of reply chains	Sustained reciprocal exchange	null	↑	Replies deepen dyadic exchanges without integrating the room
Network Density	Observed ties vs. possible ties	Group-level cohesion	↓	null	Reactions thin networks; replies leave cohesion unchanged
Diversity					
Viewpoint Diversity	Shannon Diversity	Distribution of stances	↑	↓	Reactions broaden ideological exposure; replies narrow stance range
Topic Diversity (Breadth)	Topic Entropy	Range of topics discussed	↓	null	Reactions encourage convergence on fewer topics
Topic Diversity (Distinctiveness)	KL Divergence	Uniqueness of an individual's topics vs. room average	null	↑	Replies elicit distinctive contributions without broadening stances
Structural Diversity	Betweenness/ Structural Virality	Spread across conversational subgroups	↑	null	Reactions promote broader structural reach

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concerns the 273 participants who entered a chatroom but did not post any messages. Our measures of internal inclusiveness—participation, engagement, and diversity—are defined entirely through observable behavioral traces on the platform: message volume, reply ties, stance expression, and topic distributions. Participants who did not contribute any messages, therefore, provide no data for these constructs and cannot be meaningfully included without introducing structurally inflated zeros that would distort all outcomes. Their absence from the analytic sample reflects a conceptual boundary of the present study: We focus on internal inclusiveness, or how discussions unfold among contributors, rather than external exclusiveness, or who is deterred from participating at all. The latter is an important but distinct dimension of inclusiveness that we plan to examine in future work using designs tailored to understanding entry barriers, deterrents, and silent participation.

The study design and selected hypotheses concerning linguistic deliberative quality were pre-registered; however, treatments yielded no significant effects on those outcomes. Accordingly, the present analysis focuses on inclusivity as a complementary dimension of deliberative quality. We used a single reaction variable that combines upvotes and downvotes, but future work with larger samples and greater statistical power could examine these behaviors separately. We found no interactive effects between our treatment conditions. Nevertheless, our design may have been underpowered for such an analysis. Small chatroom size (five users per room) may also explain the absence of effects on closeness centrality and triadic closure. In small groups, participants can easily address others directly without a reply feature, meaning the affordance's benefits are more apparent at larger scales (Lin et al., 2021; Velasquez, 2012). Our finding that replies increased conversation depth, but not other engagement metrics, aligns with this interpretation. Future work should test whether these dynamics scale differently in larger groups or naturally occurring discussions.

Finally, our design allowed us to isolate feature effects, but does not account for their interplay with algorithms that may use these signals to determine the prominence of posts in a social media feed (Jaidka et al., 2023). Future work should explore how interactivity combines with algorithms, identity signaling, and moderation to shape inclusiveness. Given evidence that engagement can trigger algorithmic amplification ("hot streaks") (Garimella & West, 2019; Gurjar et al., 2022) and that bots can significantly alter discussion dynamics (Pfeffer et al., 2023), understanding how affordances operate in these contexts remains a critical next step.

Conclusion

Interactivity is a bundle of distinct feature–behavior pathways, each with competing implications for inclusiveness. Experimentally isolating reactions and replies provides causal evidence that different interactive features shape participation, engagement, and diversity in systematically divergent ways.

Our study shows that interactive features introduce systematic trade-offs across different dimensions of inclusiveness. Reaction tools broaden exposure to varied stances yet suppress direct participation and concentrate attention on highly visible contributions. Reply tools support deeper exchanges and thematic elaboration but create insular conversational clusters that constrain ideological breadth. These findings challenge the assumption that adding interactivity uniformly enhances discussion quality. Instead, they underscore that the consequences of a feature depend on which dimension of inclusiveness—participation equity, contextual depth, or cross-cutting exposure—a platform seeks to advance.

These insights carry practical and normative implications. They offer guidance for designing environments that foster inclusive and meaningful discussion, but they also highlight risks. The same affordances that deepen engagement may be repurposed to privilege certain viewpoints, amplify persuasive or polarizing content, or suppress dissent. Understanding which features drive elaboration or clustering can enable beneficial interventions, but it may also encourage optimization strategies that exacerbate inequities. We therefore emphasize the need for transparency in how platforms apply experimental evidence to steer user behavior. To support accountability and replication, we will release our experimental platform for broader use.

Finally, the relevance of our findings extends beyond public feeds. As political talk increasingly migrates into private messaging, small-group exchanges, and server-based chat environments, lightweight and high-effort affordances play an even greater role in determining whose contributions surface and how conversational clusters form. Our framework helps anticipate how design decisions will shape inclusiveness in these emerging communicative spaces and provides a foundation for future work examining how interactivity operates across contexts, populations, and platform architectures.

Supplementary material

Supplementary material is available at *Journal of Computer-Mediated Communication* online.

Data availability

The dataset and scripts used in reporting the results are available at https://osf.io/x6cf9/.

AI use

Claude (Sonnet 4.6) and ChatGPT (5.2) were used to proofread a full draft of the manuscript and format the tables. All ideas, analyses, interpretations, and mistakes are the authors' own.

Code and data

Code and data for this project is available for double-blind peer review at https://osf.io/x6cf9/. The Trivium software that was custom-developed to host and design experiments is documented at http://trivium.smolproject.com/

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Conflicts of interest

None declared.

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Open science framework badges

Open Materials

The components of the research methodology needed to reproduce the reported procedure and analysis are publicly available for this article.

Open Data

Digitally shareable data necessary to reproduce the reported results are publicly available for this article.

Preregistered

Research design was preregistered.

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